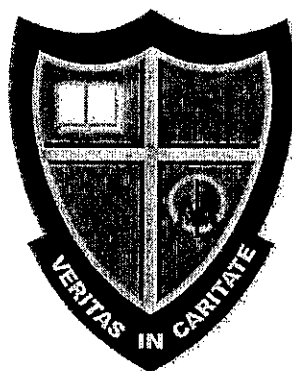


ST BENEDICT'S COLLEGE



SUBJECT	Advanced Programme Mathematics	DATE	November 2014
GRADE	11	MARKS	180
EXAMINER	Mrs Povall	MODERATOR	Mrs Eckert Mr Gill
NAME		DURATION	2 hours
CLASS			

QUESTION NO.	ASS STANDARD DESCRIPTION	TOTAL	ACTUAL
1	Solving Equations	27	
2	Solving Inequalities	10	
3	Absolute values and Absolute Value Graphs	23	
4	Complex Numbers	25	
5	Function notation and Inverses	16	
6	Split Graphs and Discontinuity	34	
7	Radian Measure and Trigonometry	29	
8	Limits and Differentiation	16	
	TOTAL	180	

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of 7 pages.
2. Read the questions carefully.
3. Answer all the questions.
4. Number your answers as the questions are numbered.
5. All the necessary working details must be clearly shown.
6. Approved non-programmable calculators may be used except where otherwise stated.
7. It is in your own interest to write legibly and to present your work neatly.
8. Where necessary, round off to two decimal places.

QUESTION 1**27 MARKS**Solve for x , where $x \in \mathbb{R}$:

a) $\frac{x-2}{x^2-x-6} = \frac{3}{2x+4} - \frac{x}{4-x^2}$ (10)

b) $1 - \frac{4}{\sqrt{x+2}} = \frac{32}{x+2}$ (6)

c) $x^2 - x = 14 - \frac{24}{x(x-1)}$ (8)

d) $x^2 + 9 = 0$ where $x \in \mathbb{C}$ (3)

QUESTION 2**10 MARKS**

Solve each of the following inequalities:

a) $\frac{x^2+4}{x-2} < 0$ (3)

b) $\frac{3-x}{x^2+9x-10} \geq 0$ (7)

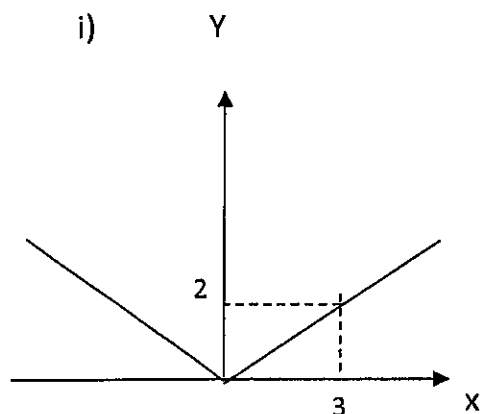
a) Solve for x : i) $|x + 1| = -2$ (1)

ii) $|2x + 1| = 4$ (2)

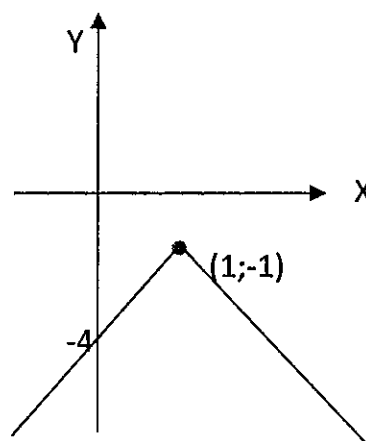
iii) $|x - 3| + 2x = -2$ (4)

b) Determine the equations of the following graphs:

i)



ii)



(6)

c) Let $f(x) = |x + 1|$ and $g(x) = -\frac{x}{2} + 1$

i) Sketch the graphs of f and g , labelling all intercepts, etc. (6)

ii) From your graph, find the values of x such that $f(x) = g(x)$ (2)

iii) From your graph determine for which values of x : (2)

$$f(x) < g(x)$$

QUESTION 4**25 MARKS**

a) Simplify and show all your working out. Give your answers in standard $a + bi$ form.

i. $(2 + 3i)(4 - 5i)$ (3)

ii. $\frac{5+5i}{2-i}$ (4)

b) Given $p(x) = 2x^3 + ax^2 + bx - 15$ with a zero at $x = 2 - i$.

Determine the value of a and b . (8)

c) i) Write down a quadratic equation, one of whose roots is $x = 2 - 3i$. Give your

answer in the form $ax^2 + bx + c = 0$ (6)

ii) Hence, or otherwise, solve $x^3 - 9x^2 + 33x - 65 = 0$ (4)

QUESTION 5**16 MARKS**

The functions f , g and h are defined for real values of x by $f(x) = |x|$;
 $g(x) = 3x + 5$ and $h(x) = g(g(x))$.

a) Solve the equation $g(x + 2) = f(-12)$ (5)

b) Find $h^{-1}(x)$ (6)

c) Determine the values of x for which $x + f(x) = 0$ (5)

QUESTION 6**34 MARKS**

a) Draw a sketch graph of $f(x)$. Label all co-ordinates of points where the domain changes if

$$f(x) = \begin{cases} 2^{-x} & \text{if } x \leq 1 \\ 0,5x & \text{if } 1 < x < 4 \\ 0 & \text{if } x = 4 \\ 6 - x & \text{if } 4 < x \leq 6 \\ (6 - x)^2 - 1 & \text{if } 6 < x \leq 8 \end{cases} \quad (10)$$

Now, for the above function:

b) Calculate $f(1)$, $f(2)$, $f(4)$ and $f(5)$ (4) ☐

c) For which values of x is $f(x) = 0$? (4)

d) Calculate $f(f(6))$ (4)

e) What are domain and range of f ? (6)

f) What type of discontinuities occur at $x=4$ and $x=6$? (6) ☐

QUESTION 7**29 MARKS**

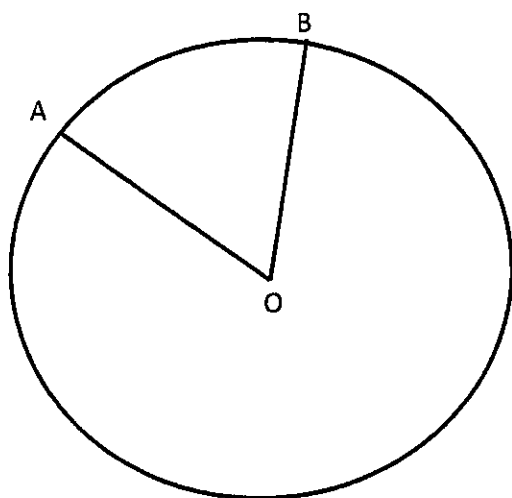
a) Convert 120° to radians. (2)

b) Convert $\frac{7\pi}{6}$ radians to degrees. (2)

c) Determine the general solution to $\sec^2 x - \sec x - 2 = 0$ in radians. (8)

d) Prove that $\frac{1}{\tan x \sqrt{1 - \cos^2 x}} = \operatorname{cosec} x \cdot \cot x$ (8)

e)



Points A and B lie on the circle with centre O.
The area of the minor sector AOB is 72cm^2
while perimeter is 36cm .

Calculate the radius of the circle.

(9)

QUESTION 8

16 MARKS

a) Evaluate: $\lim_{x \rightarrow 4} \frac{x^2 - 16}{4 - x}$ (4)

b) Evaluate: $\lim_{x \rightarrow \infty} \frac{5 - 3x - 2x^2}{(2x - 3)^2}$ (6)

c) If $f(x) = 2x^2 + 3x - 1$, Determine $f'(x)$ from first principles (6)





GRADE 11 AP MATHS PAPER 1 DATA SHEET

General Formulae

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$z = a + bi$$

$$z^* = a - bi$$

Calculus

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Trigonometry

$$A = \frac{1}{2} r^2 \theta$$

$$s = r\theta$$

$$\text{In } \triangle ABC: \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2} ab \sin C$$

$$\sin^2 A + \cos^2 A = 1$$

$$1 + \tan^2 A = \sec^2 A$$

$$1 + \cot^2 A = \operatorname{cosec}^2 A$$